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PRACTICE Series

NEET UG 2027

Zoology High-Scoring Concepts, Chapters, Mock Tests & Preparation Guide

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About the eBook

Preparing for the NEET exam requires a strategic approach, especially when it comes to mastering zoology. This eBook is your complete guide to excelling in the zoology section of NEET. It has been carefully designed to provide you with:

Comprehensive Study Material: Covering every chapter and concept in the NEET syllabus for zoology, this eBook ensures a detailed understanding of fundamental and advanced topics.

Chapter-Wise Analysis: Each chapter is broken down into bite-sized, easy-to-digest segments, focusing on conceptual clarity and application-based learning.

Important Topics Highlighted: Stay focused with a list of critical topics frequently asked in past NEET exams, along with tips to prioritise them during your preparation.

Practice Questions: Test your understanding with a wide variety of MCQs designed to simulate NEET-level difficulty.

Quick Revision Notes: Perfect for last-minute preparation, these notes summarise key points for rapid revision.

Insights from Previous Years' Papers: Get an edge by analysing trends and types of questions asked in previous NEET exams.

Whether you're revising or starting afresh, this eBook is tailored to guide you every step of the way to a top score in the zoology section.

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A Note to the Students

Dear Students,

The journey to cracking NEET is as challenging as it is rewarding. As you immerse yourself in preparation for the NEET exam, remember that consistency, focus, and a positive mindset are your greatest allies.

- **Plan Wisely:** Stick to a well-structured schedule that balances studying, practice, and self-care.
- **Master the Basics:** Understanding core physiological and structural concepts is crucial for excelling in zoology. Don’t hesitate to revisit foundational topics.
- **Practice, Practice, Practice:** Solve as many questions as you can. It strengthens your grasp of the subject and boosts confidence.
- **Use Resources Wisely:** This eBook is your tool to navigate zoology effectively – make the most of it.
- **Stay Motivated:** Challenges are inevitable, but persistence will lead you to success. Surround yourself with positivity and believe in your potential.

We wish you the very best for your preparation and are confident that with dedication and hard work, success will be yours.

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Zoology for NEET 2027: Study Material, Mock Test, Chapters, Important Topics & More

"Zoology" comes from the Greek words *"zoon"* (animal) and *"logos"* (study), and refers to the scientific study of animals, including their structure, physiology, classification, behaviour, evolution, and distribution. Aristotle is regarded as the Father of Zoology for his pioneering classification of animals. Zoology is the branch of biology that examines animal life from single-celled organisms to complex vertebrates, and it forms roughly half of the Biology syllabus tested in NEET. Zoology is important for students from Class 6 to Class 12 in the CBSE syllabus and helps students prepare for competitive exams like NEET.

Zoology Definition

Zoology is the branch of biology that deals with the scientific study of animals, their structure, physiology, classification, behaviour, distribution, and evolutionary relationships. Zoology is studied from microscopic protozoans to large mammals, explaining animal diversity and the interdependence of animal life with the environment.

What is Zoology?

Zoology is a branch of biology involving the scientific investigation of animal life. In different aspects of animal biology, zoology covers classification, anatomy, physiology, reproduction, and the interaction of animals within their ecosystems. The term “zoology” originates from the Greek words for “animal” and “study”, reflecting its early focus on documenting and classifying animal forms. Modern zoology has expanded to cover genetics, animal behaviour (ethology), conservation biology, and biotechnology applications in animal science. Broadly, zoology meaning is the scientific study of animal life, ranging from single-celled protozoa to complex vertebrates.

History of Zoology

The study of animals traces its roots back to ancient civilisations, where humans depended on animals for food, labour, and materials. Over time, this dependence turned observation into scientific investigation. Ancient scholars such as Aristotle, the Father of Zoology, laid the foundation of this branch of biology through systematic classification and description of animal species.

Early Animal Explorers

- **Aristotle:** Often called the Father of Zoology, his work *Historia Animalium* classified over 500 animal species based on observable characteristics, laying the groundwork for taxonomy.
- **Carl Linnaeus:** The Swedish naturalist introduced the binomial nomenclature system in the 18th century, giving every organism a standardised two-part scientific name that is still used today.

New Advances in the Science of Zoology

The invention of the compound microscope enabled scientists such as Antonie van Leeuwenhoek to observe microorganisms and cellular structures for the first time, revolutionising the study of animal physiology. Charles Darwin's theory of evolution by natural selection, published in 1859, transformed zoology by explaining how species change over time and how all animal life is interconnected through common ancestry.

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Branches of Zoology

Zoology is a wide field comprising several specialised branches, each concentrated on some particular aspect of animal life. Some of the main streams in zoology are as follows:

Branch	**Description**
Entomology	The study of insects, their classification, physiology, and ecological roles.
Ornithology	Focuses on the study of birds, their behaviour, migration patterns, and physiology.
Ichthyology	Examines fish species, their anatomy, classification, and habitats.
Herpetology	Deals with the study of reptiles and amphibians.
Mammalogy	Investigates mammals, including their physiology, behaviour, and evolutionary history.
Animal Physiology	Studies the internal functioning of animal organ systems, such as digestion, respiration, and circulation.
Palaeozoology	Focuses on the study of fossilised animals to understand their evolutionary history.
Ethology	The scientific study of animal behaviour, including instinct, learning, and social interaction.

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Important Parts of Zoology

Given below is the list of important parts of zoology based on the general areas that make up a typical zoology syllabus:

Animal Diversity and Taxonomy

Animal diversity and taxonomy is the study of animal classification. It includes the major animal phyla, from simple invertebrates to complex chordates, and explains how organisms are identified, classified, and named according to their evolutionary relationships and structural characteristics.

- Animal Kingdom
- Structural Organisation in Animals

Human Physiology

Human physiology explains how the major organ systems of the body function – digestion, breathing, circulation, excretion, movement, and neural and chemical coordination. This unit carries some of the highest weightage in the NEET Zoology section.

- Breathing and Exchange of Gases
- Body Fluids and Circulation & the Cardiac Cycle
- Excretory Products and their Elimination & Urine Formation and Osmoregulation
- Locomotion and Movement
- Neural Control and Coordination & Neurons and Nerves
- Chemical Coordination and Integration

Reproductive Biology

Reproductive biology covers the anatomy and physiology of the human reproductive system, gametogenesis, hormonal regulation, and reproductive health – a consistently high-scoring area in NEET Zoology.

- Human Reproduction & the Human Reproductive System
- The Menstrual Cycle
- Reproductive Health

Genetics and Evolution

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This part of zoology explains how traits are inherited across generations and how animal species have changed over geological time through natural selection, adaptation, and speciation.

- Evolution

Immunology and Human Health

This section deals with disease-causing organisms, the human immune system, and disorders affecting the body, along with the biomolecules that make up living tissue.

- Human Health and Disease
- Immunity
- Biomolecules

Applied and Economic Zoology

Applied zoology includes the study of animals with economic and medical importance, such as apiculture (bee-keeping), sericulture (silk production), animal husbandry, and the use of animal models in biotechnology and pharmaceutical research.

Practical Modules

Practical modules in zoology include laboratory dissections and specimen study, microscopy of animal tissues and cells, and field-based observation of animal behaviour and habitats. These modules reinforce the theoretical concepts taught in the NCERT textbooks and are frequently the source of diagram-based NEET questions.

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Importance of Zoology

The importance of zoology extends well beyond the classroom, contributing significantly to human health, food security, and the environment.

What does Zoology contribute to?

1. Medical and Health Sciences: Knowledge of animal physiology and immunology underpins modern medicine, from understanding disease mechanisms to developing vaccines and treatments.

2. Agriculture and Animal Husbandry: Zoology helps improve livestock breeding, dairy production, poultry farming, and pest management in agriculture.

3. Wildlife Conservation: Monitoring of animal populations and habitats by zoologists helps protect endangered species and maintain biodiversity.

4. Biotechnology and Genetic Research: Zoological knowledge supports research in genetic engineering, animal cloning, and the development of animal-model-based drug trials.

5. Ecological Balance: Understanding predator-prey relationships and food webs helps maintain the health of ecosystems and informs environmental policy.

News on Zoology

Here are some recent developments in zoology-related research in India:

- **709 New Animal Species Added to India's Faunal Database**
  The Zoological Survey of India (ZSI) announced in 2026 that 709 new animal species were added to India's official records in 2025, including 483 species entirely new to science, taking the country's documented faunal diversity past 1,05,000 species.
- **Two New Marine Worm Species Discovered**
  ZSI researchers identified two previously unknown species of marine polychaete (ragworm) from India's coastal waters, adding to the country's record of marine invertebrate diversity.
- **New Earthworm Species from the Western Ghats**
  Two new earthworm species were discovered in the Western Ghats, taking the number of known moniligastrid earthworm species in India to 95 and highlighting the region's importance as a biodiversity hotspot.

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- **Fauna of India Checklist Portal**
  ZSI launched a first-of-its-kind digital portal documenting all recorded animal species in India, aiding conservation planning and taxonomic research nationwide.

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Zoology for NEET 2027

Zoology makes up half of the Biology section of NEET, contributing 45 questions worth 180 marks out of the total 720 marks – exactly the same weightage as Botany. For NEET preparation, covering the syllabus from the NCERT textbooks is essential. Understanding key physiological concepts and practising past questions can significantly improve performance in Zoology for NEET.

Note: As per the latest reports, NEET 2027 is expected to retain the 180-question, 720-mark format (+4 for a correct answer, –1 for an incorrect answer, 0 for unattempted) and may move to Computer-Based Test (CBT) mode instead of the traditional pen-and-paper format. This change has not yet been confirmed by NTA and will be updated once the official NEET 2027 information bulletin is released.

NEET Zoology Preparation

For effective preparation of Zoology for NEET 2027, understanding the NEET Zoology syllabus is the first and most important step. The syllabus for NEET Exam 2027 covers the chapters of CBSE Class 11 and 12. Having command of the topics and concepts of Zoology classes 11 and 12 becomes the top priority for aspiring students. Given below is the chapter-wise weightage of NEET Zoology questions asked over the last five years:

Chapter name	2022	2023	2024	2025	2026	Total
Animal Kingdom	4	4	6	6	4	24
Human Health and Disease	4	4	4	7	5	24
Human Reproduction	4	3	4	7	5	23
Biomolecules	3	3	5	6	2	19
Chemical Coordination and Integration	0	2	4	6	3	15
Structural Organisation in Animals	4	5	2	3	1	15
Body Fluids and Circulation	2	1	2	0	4	9
Locomotion and Movement	3	2	2	0	2	9

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Reproductive Health	2	2	2	0	1	7
Neural Control and Coordination	0	2	1	0	2	5
Excretory Products and their Elimination	0	2	2	1	1	6
Breathing and Exchange of Gases	3	1	2	0	1	7

Source: Careers360 5-year NEET Zoology question-paper analysis (2022–2026). Figures denote the number of questions asked from each chapter in that year's NEET paper.

Animal Kingdom, Human Health and Disease, and Human Reproduction are consistently the highest-weightage Zoology chapters and should be prioritised first, followed by Biomolecules and Chemical Coordination and Integration.

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Class 11 Zoology Chapters for NEET 2027

Given below is the list of important Zoology chapters from the Class 11 syllabus for NEET Exam preparation:

Animal Kingdom	Structural Organisation in Animals
Biomolecules	Breathing and Exchange of Gases
Body Fluids and Circulation	Excretory Products and their Elimination
Locomotion and Movement	Neural Control and Coordination
Chemical Coordination and Integration	

Also Read:

- NCERT solutions for class 11 Biology
- NCERT exemplar solutions for class 11 Biology

Class 12 Zoology Chapters for NEET 2027

Given below is the list of important Zoology chapters from the Class 12 syllabus for NEET Exam preparation:

Human Reproduction	Reproductive Health
Human Health and Disease	Evolution

Note: Genetics-focused chapters that are common to both Botany and Zoology (Principles of Inheritance and Variation, Molecular Basis of Inheritance) and ecology chapters (Organisms and Populations, Ecosystem) are covered in detail in the companion "Botany for NEET 2027" eBook.

Also Read:

- NCERT exemplar solutions for class 12 Biology
- NCERT solutions for class 12 Biology

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Tips for Zoology Preparation

Here are some tips to prepare Zoology for the NEET Exam:

- **Focus on the NCERT Textbook:** Study the NCERT textbook thoroughly, as most NEET questions are asked directly from NCERT lines and diagrams.
- **Understand Physiology Concepts:** Build a strong understanding of human physiology, since it carries the highest weightage within Zoology.
- **Make Notes:** Prepare notes on important points, flowcharts, and diagrams for quick revision.
- **Practice Diagrams:** Zoology involves many labelled diagrams (nephron, heart, neuron, reproductive organs) – practise them regularly.
- **Solve Previous Papers:** Regularly solve NEET and CBSE previous year questions to understand recurring patterns.

Also read: NEET Biology Previous Year Papers

Best Zoology Books for the NEET Exam

The best Zoology books for NEET preparation are:

- NCERT textbook Biology Class 11
- NCERT textbook Biology Class 12
- Trueman's Elementary Biology
- Biology by Campbell and Reece
- MTG's NCERT at Your Fingertips
- Dinesh Objective Biology

All these books provide students with huge collections of MCQs, explanations, and practice papers for improvement in Zoology NEET preparation.

Zoology MCQ for NEET

Practice is very important in Zoology for the NEET exam, since it gives clarity on concepts of human physiology, reproduction, and animal diversity. Regular practice of similar question types increases familiarity with exam patterns and helps ensure accuracy and speed. Proper revision and focused practice regularly help students score well in the NEET exam.

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Practice MCQ On NEET Zoology	Download PDF

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Additional Resources

Students can also download the NEET Mock Test 2027 for all subjects from the links given below:

Title	Download PDF
NEET Mock Test Zoology 2027 PDF	Download
NEET Mock Test Biology 2027 PDF	Download
NEET Mock Test Chemistry 2027 PDF	Download
NEET Physics Mock Test 2027 PDF	Download

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Other Useful Resources

Some useful resources for NEET 2027 preparation are listed below:

Zoology for NEET 2027 – High-Scoring Concepts

Want to score higher in NEET Biology? Zoology carries significant weight in the NEET exam, and mastering key physiology chapters can make a big difference. This eBook compiles high-scoring concepts, important chapters, and practice mock tests to sharpen your understanding.

» Download now and master Zoology for NEET!

NEET Zoology: Most Important Chapters and Concepts – 5-Year Analysis

This detailed Careers360 analysis breaks down the highest-weightage NEET Zoology chapters based on five years of previous papers, along with a month-by-month preparation strategy.

» Read the full 5-year Zoology analysis!

NEET Biology Sample Paper 2027 (Latest Pattern + Full-Length Practice PDF)

Preparing for NEET 2027 and want to strengthen your Biology score with real exam-level practice? This NEET Biology Sample Paper 2027 PDF simulates the actual exam experience with a balanced mix of questions from Human Physiology, Genetics & Evolution, Ecology, Biotechnology, and Plant Physiology.

» Download the NEET Biology Sample Paper 2027 PDF now!

Top 100 NEET Physics Questions 2027 (Concept-Based & Numerical) – PDF with Solutions

A carefully curated collection of concept-based and numerical Physics questions with step-by-step solutions, focused on high-weightage topics like Mechanics, Electricity, Modern Physics, and Thermodynamics.

» Download the Top 100 NEET Physics Questions PDF!

Top 100 NCERT-Based Questions for NEET 2027 (Physics, Chemistry & Biology)

A specially curated set of most-expected questions across all three subjects, designed strictly on NCERT concepts and previous-year trends, with step-by-step explanations.

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» Download the Top 100 NCERT-Based Questions PDF!

Top 100 NEET Biology Questions to Score 330+ Marks

Carefully selected questions from Human Physiology, Genetics & Evolution, Ecology, Biotechnology, and Plant Physiology, based on previous-year trends, with clear explanations.

» Download now and target 330+ marks in NEET Biology!

NEET Biology NCERT Line-by-Line Questions PDF

Master every concept directly from NCERT with carefully framed line-by-line questions and detailed answers to improve accuracy and conceptual clarity.

» Download now to boost your NEET Biology score!

NEET Human Physiology PYQs (100 Questions with Solutions)

This PYQ collection covers Digestive, Circulatory, Nervous, Respiratory, Excretory, and Endocrine systems – the core of NEET Zoology – with detailed solutions for each question.

» Download now and sharpen your Zoology concepts!

NEET Genetics & Evolution PYQs (100 Questions with Solutions)

This resource compiles 100 previous year questions with step-by-step solutions, ideal for mastering the high-weightage Genetics and Evolution topics of NEET Zoology.

» Download here and solve Genetics & Evolution PYQs!

NEET Biotechnology & Its Applications PYQs (100 Questions with Solutions)

Biotechnology is an important unit that overlaps with Zoology through genetic engineering and biopharmaceuticals. This resource offers 100 carefully selected previous year questions with detailed solutions.

» Download now and strengthen Biotechnology for NEET!

NEET Class 11 Biology Important Diagrams PDF

Diagrams are an essential part of NEET preparation, especially the labelled diagrams of the nephron, heart, neuron, and reproductive systems from Class 11 Zoology.

» Download now and revise NEET Biology diagrams effectively!

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Biology Mind Maps for NEET 2027

Simplify complex concepts with this mind map collection covering all key topics across Botany and Zoology, including Human Physiology, Genetics & Evolution, and Ecology.

» Download now and sharpen your Biology concepts!

NEET Ultimate 60-Day Study Plan – Crack NEET in 2 Months

This free study plan shows you how to prepare for NEET effectively in just 60 days, with a daily schedule, chapter priorities, and smart revision tips.

» Download now and follow the NEET 60-Day Study Plan!

NEET Most Scoring Chapters & Topics (Past 5-Year Analysis)

This eBook analyses NEET papers from the past five years and highlights the most scoring chapters in Physics, Chemistry, and Biology, helping you prioritise your revision.

» Download here and prepare with NEET's most scoring chapters!

NEET Previous 10 Years Questions with Detailed Solutions

This eBook covers the last 10 years of NEET questions across Biology, Physics, and Chemistry with detailed, step-by-step solutions – ideal for identifying question trends.

» Download now and master previous years' papers!

NEET Formula Sheet 2027 – Physics & Chemistry

A free formula sheet compiling all the important Physics and Chemistry formulas for NEET 2027 in one place – perfect for last-minute revision.

» Download here and revise with the NEET 2027 Formula Sheet!

NEET Important Formulas – Chemistry

This eBook compiles all the important formulas for Physical, Organic, and Inorganic Chemistry that are frequently asked in NEET.

» Download now and memorise key Chemistry formulas for NEET!

NEET Important Formulas – Physics

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This free PDF brings you all the essential formulas from Mechanics, Optics, Electricity, Magnetism, and Modern Physics to help you solve numerical problems faster.

» Download here and master Physics formulas for NEET!

NEET Chemistry Cheat Sheet – Must-Know Formulas for Quick Revision

A crisp, easy-to-revise formula cheat sheet for Physical, Organic, and Inorganic Chemistry – your perfect last-minute revision guide.

» Download now and revise Chemistry with the NEET Cheat Sheet!

NEET Botany Mock Test 2027

Focused practice matters! This mock test PDF is designed exclusively for Botany, the companion subject to Zoology within the NEET Biology section.

» Download now and solve the NEET Botany Mock Test 2027!

Botany for NEET 2027: High-Scoring Concepts, Chapters, Mock Tests & Preparation Guide

The companion eBook to this one – covering the Botany half of the NEET Biology syllabus in the same chapter-wise, high-scoring format.

» Download the Botany for NEET 2027 eBook!

NEET Previous 5 Years Question Papers with Solutions (2021–2025)

One resource, all papers! This PDF compiles NEET question papers from the last five years, complete with detailed solutions – a must-have for serious aspirants.

» Download now and access NEET's previous 5-year solved papers!

NEET 2026 Free Mock Test (PDF) with Solutions

A full-length NEET mock test simulating the real exam experience, with detailed solutions to help you evaluate your preparation level before NEET 2027.

» Download now and attempt the NEET Free Mock Test!

NEET 2026 Answer Key with Solutions (All Codes)

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Practising with an official-pattern question paper and answer key is one of the most effective ways to analyse your performance ahead of NEET 2027.

» Download now and practice with the NEET 2026 Answer Key!

All resources above are from the Careers360 Medicine Prep library and open directly on medicine.careers360.com, school.careers360.com, or www.careers360.com. Titles marked as "2027" editions will be refreshed automatically once NTA releases the official NEET 2027 bulletin.

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